

Reproducible Experiments

Project: Responsible Plant Classifier

1. Reproducible Experiments

Teach the learning cycle

Primary target: Record enough detail to reproduce a run

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Metrics Match Consequences

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a declared score threshold including equality

Reference example, not the only acceptable wording: Teacher explanation: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

Hold Out Test Data

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

Look for: Freeze the full evaluation procedure before testing

Reference example, not the only acceptable wording: Teacher explanation: Accuracy 15/20, precision 3/5, recall 3/6. Report TN 12 FP 2 FN 3 TP 3 and blocked positive.

Mean, Median, and Baselines

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

Look for: Fit mean and median constant baselines on training targets

Reference example, not the only acceptable wording: Teacher explanation: Mean 6; median 5.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Two teams use the same random seed but different label files. Their results differ. List the records needed to reproduce the experiment.

Concept to explain

A result is reproducible only when another builder can recreate the data, split, code, settings, seed, and metric calculation.

Term · Meaning

Term: reproducibility

Meaning: ability to recreate a result

Term: data version

Meaning: identified dataset state

Term: random seed

Meaning: value controlling repeatable randomness

Term: experiment record

Meaning: complete inputs, settings, and outcomes

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on experimental records and seeds. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer. Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

The model version, split, metric, data provenance and development choices.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank evidence table

Saved classroom project records for critique; no personal student dataset required

Apply the connected idea: Versioned experiments

Teach Week 28 as a two-session investigation using fictional data. A fictional activity scoring experiment compares two weight settings on the same development records. Save code version, input version, weights, tie rule and outputs. A label such as v2 is useful only when it refers to an identifiable saved state. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

To avoid changing expectations simply to match a bug.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can explain reproducible experiments using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Reproduce a peer's run from an experiment card.

MODEL IDEA A result is reproducible only when another builder can recreate the data, split, code, settings, seed, and metric calculation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Record data=v2, split=grouped-s1, code=commit A7, seed=42. Record preprocessing, features, model type, settings, and library versions.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Record data=v2, split=grouped-s1, code=commit A7, seed=42.

STEP 2 Record preprocessing, features, model type, settings, and library versions.

STEP 3 Peer reruns and gets 16/20 correct, matching 80% accuracy.

STEP 4 A mismatch triggers environment and data comparison—not silent replacement.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

A report says accuracy85% without data split, code version, seed or preprocessing.

Specify reproducibility records.

Exact explanation to model

Record data identity, group/split IDs, code/dependency versions, preprocessing, features, settings, seeds where relevant, metric definition and raw predictions. Seed alone is insufficient.

Explain the added worked example

A seed can help repeat a random split or initialization, but it is not a magic quality stamp. Save data version, split membership, code version, settings, preprocessing, seed and metric definition. Some software operations remain nondeterministic, so document tolerances and environment too.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines experimental records and seeds.

A team declares seeds 11, 22, 33 before running. Validation accuracies are 14/20, 16/20, 12/20:70, 80, 60 percent. Mean accuracy is 70 percent, with a 20-percentage-point range. Reporting only seed 22 as the model result hides the declared experiment.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Key terms

Term · Meaning

Term: Seed

Meaning: An initial value controlling a pseudo-random process

Term: Run record

Meaning: A saved account of inputs, settings and outputs

Term: Reproducibility

Meaning: Ability to repeat a specified procedure

Term: Cherry-picking

Meaning: Reporting favorable results while hiding relevant others

Apply the connected idea: Versioned experiments

A fictional activity scoring experiment compares two weight settings on the same development records.

Save code version, input version, weights, tie rule and outputs. A label such as v2 is useful only when it refers to an identifiable saved state.

The ranking changes because the preference weights changed. That is a policy comparison, not proof that v2 is more accurate. Keep the same data and constraints when comparing weights, and record why a policy better fits the stated goal.

No. The data change is another possible explanation. Exit: No. They now informed development. Report that use and seek fresh cases for a new independent check.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Experimental records and seeds at a glance

Part · Evidence or action · Interpretation

Part: Before running

Evidence or action: Declare 11 22 33

Interpretation: Save data and settings

Part: Observed results

Evidence or action: 70 80 60 percent

Interpretation: Keep all three runs

Part: Summary

Evidence or action: Mean 70 percent

Interpretation: Range 20 points

The seed is part of the procedure, not evidence that a split represents the target population. Reproducing leakage perfectly reproduces a mistake. Check group separation and legitimate features before celebrating identical outputs.

A file contains only seed 22 and 80 percent. Name four missing items. Explain the steps or evidence behind your response.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Reproducible Experiments

Record variation across declared seeds instead of selecting the prettiest run.

The facts we will use

Same fixed validation set of 10 cases; declared runs seed 4=6 correct, seed 5=8, seed 6=7. All three are reported.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

All declared seeds

accuracy percent



Bar length shows the supplied value; the marked baseline is zero.

Reproducible Experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep run identities

Record seeds 4, 5, 6 with 6/10, 8/10, 7/10.



2. Summarise variation

Mean accuracy is 70%; range is 60% to 80%.



3. Preserve context

Record data version, split, code, settings and seeds, not just the winning 80%.

CHANGE ONE THING

The report deletes the 60% run.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Keep run identities

What happens and why: Record seeds 4, 5, 6 with 6/10, 8/10, 7/10.

Step: 2. Summarise variation

What happens and why: Mean accuracy is 70%; range is 60% to 80%.

Step: 3. Preserve context

What happens and why: Record data version, split, code, settings and seeds, not just the winning 80%.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Record data version, split, code, settings and seeds, not just the winning 80%. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Seeds help reproduction but do not necessarily remove hardware or library nondeterminism; report the actual environment.

Change one thing in the pictured case

Change: The report deletes the 60% run.

Prediction: The remaining mean rises to 75%, but the reported comparison becomes selectively incomplete.

Reason: Removing an unfavourable declared run changes the evidence rather than improving the method.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A report says accuracy 85% without data split, code version, seed or preprocessing.

Write the question to explain: Specify reproducibility records.

Reveal after predictions: Record data identity, group/split IDs, code/dependency versions, preprocessing, features, settings, seeds where relevant, metric definition and raw predictions. Seed alone is insufficient.

Keep for an independent attempt: Two runs use different random splits and different feature sets. Their scores differ. What must be recorded, and why is the cause unclear?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: Examples: data version, split IDs, model settings, code version, metric denominator and full planned-run results. Exit response: Record the failed run, cause and any retry; disclose which runs contributed to summaries and do not invent its score.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: What must accompany a reported validation score? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify seed in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: summarize all runs; improve a record. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between seed and run record. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Does identical output prove a valid experiment?

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Record enough detail to reproduce a run; Distinguish reproducibility from validity. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Teach the connected idea

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and explain the purpose in the opening section.

Stage: 7 to 20 minutes

Teacher action: Read the worked example and diagram; ask learners to trace each step aloud.

Stage: 20 to 33 minutes

Teacher action: Complete W1 and W2, then compare predictions with observed results.

Stage: 33 to 45 minutes

Teacher action: Collect the guided prediction before running; discuss a mismatch and its cause.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the main rule without showing the previous answer.

Stage: 8 to 20 minutes

Teacher action: Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion task: A file contains only seed 22 and 80 percent. Name four missing items.

Teacher check: Teacher explanation: Examples: data version, split IDs, model settings, code version, metric denominator and full planned-run results.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Experimental records and seeds. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Mean 70 percent; range 80-60=20 percentage points, across the three declared runs.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Versioned experiments

Target: Record a reproducible experiment version

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: A has fun 3 and learning 2. Score it under weights (2,1) and (1,2).

Teacher check: Scores 8 and 7.

Individual check: For Versioned experiments, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected correction: Once holdout results guide another change, those cases have become development evidence. A later independent evaluation needs new reserved cases. Small holdouts still provide limited evidence.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

What should accompany the mean 70%? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

The three individual run results, seed list, 60–80% range and shared evaluation conditions, so variability is visible. Changed-case check: The remaining mean rises to 75%, but the reported comparison becomes selectively incomplete. Removing an unfavourable declared run changes the evidence rather than improving the method.

Changed-case reasoning: The remaining mean rises to 75%, but the reported comparison becomes selectively incomplete. Removing an unfavourable declared run changes the evidence rather than improving the method.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Reproduction Swap: exchange incomplete experiment cards, identify missing fields, then reproduce and explain any mismatch.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Writing down the final accuracy makes an experiment reproducible.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Mean 70 percent; range $80-60=20$ percentage points, across the three declared runs.

A stronger claim makes a stronger project: Record the failed run, cause and any retry; disclose which runs contributed to summaries and do not invent its score.

The new case can be copied from the example: The independent task changes the evidence. Require a fresh explanation: Mean $27/36=75$ percent; range $(10-8)/12=1/6$ about 16.7 percentage points. Retain all three runs.

Check these misconceptions

A successful run proves the conclusion: Once holdout results guide another change, those cases have become development evidence. A later independent evaluation needs new reserved cases. Small holdouts still provide limited evidence.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: E (4,2), F (1,5) gives v_1 10 versus 7 and v_2 8 versus 11. Require a complete run log and a statement of what is held fixed.

A convenient shortcut preserves meaning: Overwriting a file called latest removes the ability to reproduce earlier results unless versions are retained elsewhere. Save identifiable versions and a run log. Avoid editing a reserved final test after seeing its output to make the chosen version look better.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Record data identity, group/split IDs, code/dependency versions, preprocessing, features, settings, seeds where relevant, metric definition and raw predictions. Seed alone is insufficient.

Independent question

Two runs use different random splits and different feature sets. Their scores differ. What must be recorded, and why is the cause unclear?

Expected independent answer

Record data version, split or seed, features, model settings, code version and metric. Since multiple factors changed, the score difference cannot be attributed to one factor.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “A random seed alone records every source of experimental variation.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Record dataset and label versions, identity groups and split IDs, code/dependency versions, preprocessing, features, model settings, seeds, metric definitions and raw predictions. Same seed cannot compensate for different data or procedures.

Explanations for the additional workbook examples

W1 Summarize all runs

Calculate the worked mean and range in percentage points.

Teacher explanation: Mean 70 percent; range $80-60=20$ percentage points, across the three declared runs.

W2 Improve a record

A file contains only seed 22 and 80 percent. Name four missing items.

Teacher explanation: Examples: data version, split IDs, model settings, code version, metric denominator and full planned-run results.

W3 Separate two ideas

Does identical output prove a valid experiment?

Teacher explanation: No. A deterministic experiment can still contain leakage, invalid labels or an unrepresentative split.

W4 Apply independently

Declared seeds 1, 2, 3 produce 9/12, 8/12, 10/12 correct. Compute the mean accuracy and range.

Teacher explanation: Mean $27/36=75$ percent; range $(10-8)/12=1/6$ about 16.7 percentage points. Retain all three runs.

W5 Handle a failed run

One planned seed fails before prediction. How should it appear in the record?

Teacher explanation: Record the failed run, cause and any retry; disclose which runs contributed to summaries and do not invent its score.

Versioned experiments — W1 Record a run

List five items needed to reproduce a scoring run.

Code version, input version, configuration weights, tie/eligibility rules and outputs; also record date and method as useful context.

Versioned experiments — W2 Compute a comparison

A has fun 3 and learning 2. Score it under weights (2,1) and (1,2).

Scores 8 and 7.

The weight pair is (fun weight, learning weight). Under (2,1), the score is $2*3 + 1*2 = 8$. Under (1,2), it is $1*3 + 2*2 = 7$. The same ratings were kept fixed; the difference comes from changing the preference policy.

Versioned experiments — W3 Repair an experiment

A learner changes weights, input ratings and the time limit together. Why is attribution difficult?

Multiple factors changed. Compare one deliberate change while holding the others fixed, or explicitly describe a multi-change experiment without attributing it to one cause.

Versioned experiments — W4 Design independently

Plan two versions for C (fun 5, learning 1) and D (fun 2, learning 5), using weights (2,1) and (1,2). Record expected scores, ranks, fixed data and a separate final-case plan.

v1 C 11, D 9 selects C; v2 C 7, D 12 selects D. Preserve the same records and other rules; reserve unseen cases until the policy is fixed.

Versioned experiments — W5 Protect evaluation

You revise the policy after reading final-case failures. Are those cases still untouched holdout evidence?

No. They now informed development. Report that use and seek fresh cases for a new independent check.

Assess independent transfer

Selected case: Declared seeds 1, 2, 3 produce 9/12, 8/12, 10/12 correct. Compute the mean accuracy and range.

Selected case answer

Teacher explanation: Mean $27/36=75$ percent; range $(10-8)/12=1/6$ about 16.7 percentage points. Retain all three runs.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Give every Plant Classifier experiment an ID and immutable record linking dataset, split, code, settings, predictions, and metrics.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 27: Overfitting. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank evidence table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use seed and run record in full sentences. Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Mean $27/36=75$ percent; range $(10-8)/12=1/6$ about 16.7 percentage points. Retain all three runs. Fresh retry accuracies 50, 60, 70 percent: mean 60 percent, range 20 percentage points. Require all declared runs and their denominators. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Distribution shift. Carry forward the saved evidence and the distinction between development choices and justified claims.

Readiness and reassessment for Versioned experiments

Provide a run-log table. Extension: rerun a saved version in a fresh folder and compare exact output values.

Independent W4 evidence: v1 C 11, D 9 selects C; v2 C 7, D 12 selects D. Preserve the same records and other rules; reserve unseen cases until the policy is fixed. Record whether the learner needed no help, a hint or explanation. Fresh retry: E (4,2), F (1,5) gives v1 10 versus 7 and v2 8 versus 11. Require a complete run log and a statement of what is held fixed.

Plan the delayed return

Suggested return points in this level: ai-4/27, ai-4/29, ai-4/32. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

7. Prepare, teach and assess

Teaching target: Record enough detail to reproduce a run

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: A file contains only seed 22 and 80 percent. Name four missing items.

Reasoned practice answer: Teacher explanation: Examples: data version, split IDs, model settings, code version, metric denominator and full planned-run results.

Check every learner before discussion: Use the worked case for Experimental records and seeds.

Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Mean 70 percent; range 80-60=20 percentage points, across the three declared runs.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The report deletes the 60% run. Require a prediction and a reason before comparing. Expected result: The remaining mean rises to 75%, but the reported comparison becomes selectively incomplete. Reason: Removing an unfavourable declared run changes the evidence rather than improving the method.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test

and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.